

BC Housing Price Predictor

Predicting BC Property Prices with Machine Learning

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\$100K – \$5M

Range of home prices
in Greater Vancouver



21,977

BC property listings
in our dataset



81%

Variance explained
by Random Forest (R^2)

The Problem

BC's real estate market is complex and opaque. Buyers, sellers, and investors lack transparent, data-driven price estimates.

Our Solution: A supervised machine learning system that predicts housing prices from property features that helps buyers make informed decisions and gives real-estate professionals a powerful tool.

21,977

Property Records

9 → 243

Features (after encoding)

80 / 20

Train / Test Split

5

Numerical Features

2

Categorical Features

5-Step AI Pipeline:

①

Raw Data

raw_data.csv
21,977 listings

②

Preprocessing

OneHotEncoder
StandardScaler

③

Feature Eng.

9 → 243
features

④

Model Training

6 models
GridSearchCV cv=3

⑤

Evaluation

MAE, RMSE, R²
.pkl saved

1

Linear Regression

Baseline. No regularisation.

Baseline

4

Decision Tree

Non-linear splits. max_depth=10.

Tree-based

2

Ridge Regression

L2 regularisation. $\alpha=1.0$ (GridSearchCV).

Regularised

5

K-Nearest Neighbours

Distance-weighted. k=9, Manhattan.

Instance-based

3

Lasso Regression

L1 regularisation. Feature-selecting.

Regularised

6

Random Forest

200 trees ensemble. Best performer.

Ensemble

Results – Model Comparison

MAE

\$228,287

Mean Absolute Error

RMSE

\$405,301

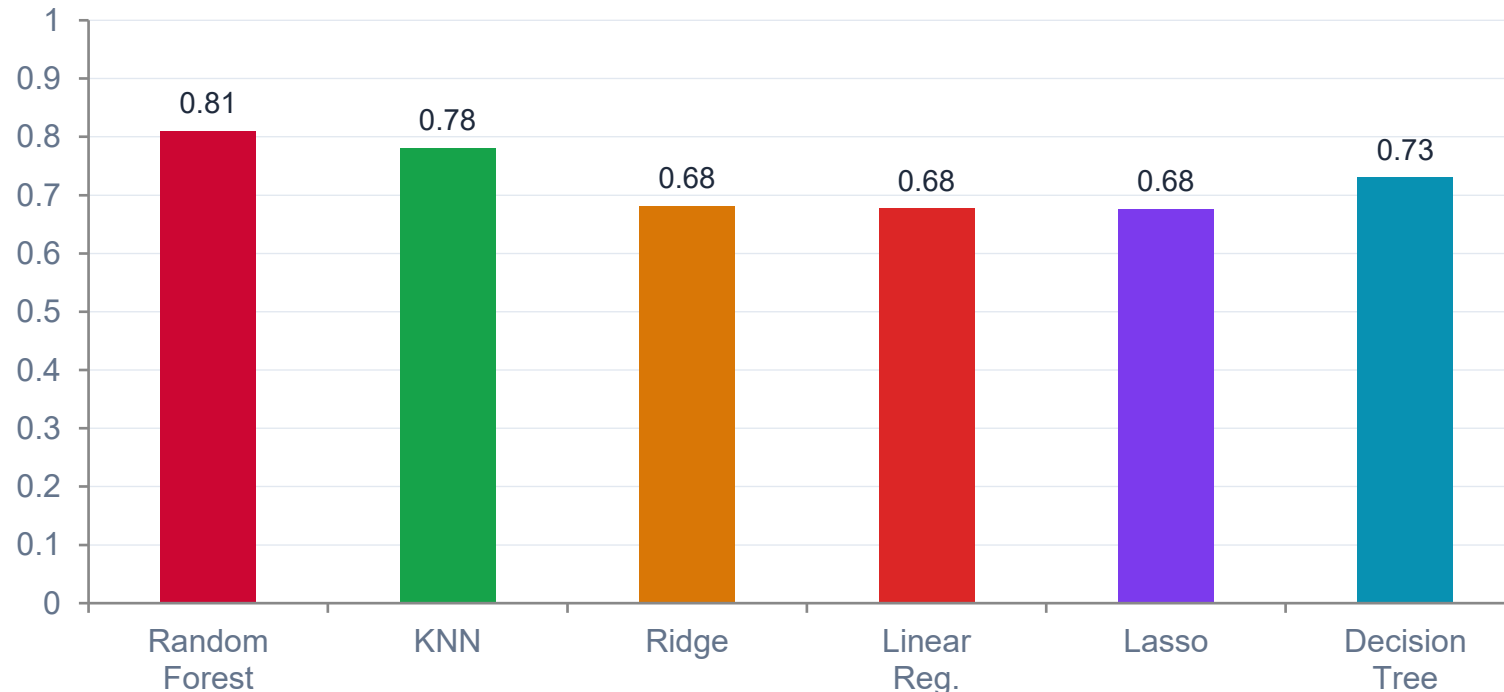
Root Mean Sq. Error

 R^2 **0.8143**

Variance Explained

**Best Model****Random Forest***Explains 81% of price variance*

R^2 Score by Model (↑ Higher is Better)



Key Insights

Random Forest $R^2=0.81$ which is best by farKNN $R^2=0.78$, solid but slowerLinear models plateau at $R^2 \approx 0.68$ 

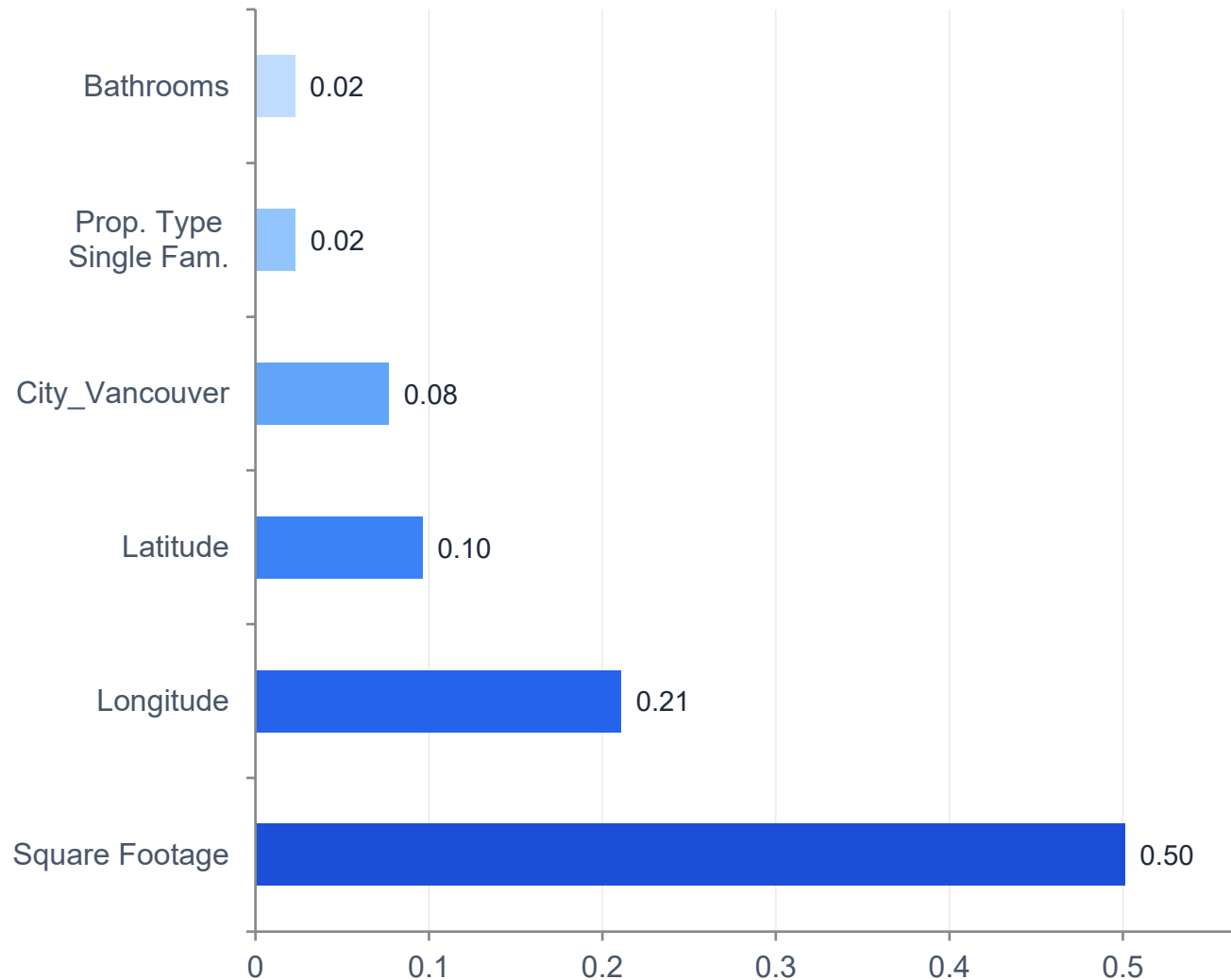
High MAE reflects BC's wide price range (\$100k–\$5M+)



5 models tuned via GridSearchCV, cv=3

Feature Importance & Key Findings

Random Forest – Feature Importance (Mean Decrease in Impurity)



Size Dominates

Square footage alone accounts for 50% of the model's importance to buyers, who pay for space.



Location Matters

Longitude + Latitude + City_Vancouver = 38% combined. Vancouver proximity commands a premium.



Ensemble Wins

Random Forest ($R^2=0.82$) outperforms all linear models ($R^2\approx0.68$), and non-linear effects are real.



Dimensionality

230+ one-hot columns hurt KNN (curse of dimensionality) and caused Lasso convergence issues.

Conclusion & Future Work

7 / 7

What We Achieved

- ✓ Built & compared 6 regression models on 21,977 BC listings
- ✓ Random Forest achieved $R^2=0.82$, MAE=\$226,350
- ✓ Identified Square Footage & Location as key price drivers
- ✓ Full GridSearchCV hyperparameter tuning pipeline
- ✓ Reproducible .pkl model pipelines saved for deployment

Thank You! Questions?

Future Work

 XGBoost & neural network models (MLP, TabNet)

 Incorporate time-series & market trend data

 Add proximity to transit, schools, walk score

 Deploy as an interactive web price estimator

 Expand dataset to include more BC regions

 github.com/ammарprасla/BC-Housing-Price-Predictor